Band-1 Hadamard Order 4388 — Miyamoto (Double Paley-I seed H_1096); Gap 2052 Still OPEN

H_548/H_1096/C_1098/H_4388 PASS; 4388 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4372_Miyamoto`; `2026-09-15_Band1_Hadamard_4196_Miyamoto` (Double Paley-I seed pattern).
Lab. `/workspace/experiments/band1/hadamard_4388_miyamoto/`.
Cutoff. 2026-09-15T15:40:20Z (~11:40AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 548, 1096 and 4388 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{1096} (H_{548} Paley-I $q = 547$, then Sylvester double). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4388 is a **Miyamoto-only bonus (NOT in SPK)**: Paley-I **FAIL** ($q = 4387 = 41 \cdot 107$ not prime); Paley-II **FAIL** ($q = 2193 = 3 \cdot 17 \cdot 43$ not prime); Kronecker **FAIL** ($4388 = 4 \cdot 1097$, 1097 not Hadamard-admissible). Seed triage for H_{1096}: Paley-II **FAIL** ($q = 547 \equiv 3 \pmod 4$, needs $\equiv 1$); Paley-I direct **SKIP** ($q = 1095 = 3 \cdot 5 \cdot 73$ not prime); Double(Paley-I) **PASS**. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4372 on box. This increment ships **order 4388** ($4388 = 4 \cdot 1097$, 1097 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Double(Paley-I) seed** H_{1096} (H_{548} from Paley-I $q = 547$, Sylvester-doubled). Symmetric Paley conference C_{1098} over prime 1097; $H_{4388} = \text{Miyamoto}(H_{1096}, C_{1098})$. Unit tests **39/39 PASS**. SPK next gap remains **2052** (OPEN). Order 4388 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4388; Miyamoto; Paley type I; Sylvester double; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{548} constructed and verified (Paley type I, $q = 547$). 2. H_{1096} constructed and verified (Sylvester double of H_{548}). Paley-II path for H_{1096} **rejected** ($q = 547 \equiv 3 \pmod 4$). 3. C_{1098} constructed and verified (Paley conference, $q = 1097$): $CC^\top = 1097I$, zero diagonal. 4. H_{4388} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $4388 \notin$ SPK (Miyamoto-only). Paley-I/II/Kronecker paths for 4388 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$4388 = 4 \cdot 1097$
$q = 1097$	prime, $1097 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1096}
Needed conference	order $1098 = q + 1$
H_{1096} path (used)	Double(Paley-I): H_{548} Paley-I $q = 547$, Sylvester double $\rightarrow H_{1096}$
H_{1096} Paley-II	**FAIL**: $2(q+1) = 1096 \Rightarrow q = 547 \equiv 3 \pmod 4$ (needs $\equiv 1$)
H_{1096} Paley-I direct	**SKIP**: $q + 1 = 1096 \Rightarrow q = 1095 = 3 \cdot 5 \cdot 73$ not prime
Conference path	Symmetric Paley over prime 1097
Paley-I for 4388	**REJECTED**: $q' = 4387 = 41 \cdot 107$ not prime
Paley-II for 4388	**REJECTED**: $q' = 2193 = 3 \cdot 17 \cdot 43$ not prime
Kronecker for 4388	**REJECTED**: only 4×1097; $1097 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{548}	Paley type I, $q = 547$	0	0.2415	**PASS**
H_{1096}	Sylvester double of H_{548}	0	0.5986	**PASS**
C_{1098}	Paley conference, $q = 1097$	0	1.1454	**PASS**
H_{4388}	Miyamoto(H_{1096}, C_{1098})	0	39.0682	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 42.2304 s. Compressed ± 1 matrices saved as `H548_pm1.npz`, `H1096_pm1.npz`, `C1098_pm1.npz`, `H4388_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior classical/SPK ships (through 4036/3908/3748/… + **1020** Paley-II + **1032**/**1092** Paley-I + **524**/**1048** Double Paley-I; **exclude** 4372/4196/4132/4084/3716/3524/3284 Miyamoto-only) $\cup$ verified H_{548} (Paley-I) $\cup$ verified H_{1096} (Sylvester-double) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4500). **Miyamoto is not folded into SPK**; order 4388 remains a Miyamoto-only bonus.

Quantity	Value
Known count	964
4388 in SPK	**False** (Miyamoto-only)
1096 in SPK	**True** (Sylvester-double of Paley-I)
548 in SPK	**True** (Paley-I)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**Yes** (like 4372/4196/4132/4084/3716/3524/…)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

39/39 PASS (Sylvester/Paley/Miyamoto controls; $q = 547/1097$ arithmetic; Double Paley-I seed for H_{1096}; Paley-II seed rejection; SPK path rejection for 4388; Miyamoto-only arith guard; onbox seeds include 504/524/1020/1032/1048/1092 / exclude 4084/4132/4196/4372/4388; reject-all-ones; conference control; gap-2052-not-filled guard).

7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 4388 is a **Miyamoto-only bonus (NOT in SPK)**.

8. Next optional

method-attack SPK gap **2052** OR next Miyamoto ($q = 1117 \to n = 4468$ PaleyII($q = 557$); or $q = 1129 \to n = 4516$ Double(PaleyI half=564); or $q = 1153 \to n = 4612$ PaleyI($q = 1151$)) OR Track B privacy / Object 1 meter await.

Lab: `/workspace/experiments/band1/hadamard_4388_miyamoto/` · *Script:* `verify_hadamard_4388.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.